An Applied Study on WPT in Denoising Torpedo Magnetic Fuze Signal

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***Abstract*—Torpedo fuze signal denoising is an important action to ensure reliable operation of fuze. Based on the good characteristics of wavelet packet transform (WPT) in signal denoising, the paper used wavelet packet transform to denoise the magnetic fuze signal under a complex background interference, and a simulation of the denoising results with Matlab is performed. Simulation result shows that the WPT denoising method can effectively eliminate background noise existed in torpedo magnetic fuze target signal with higher precision and less distortion, leading to advance the reliability of torpedo magnetic fuze operation.**

Keywords-torpedo; magnetic fuze; signal denoising; wavelet packet transform

I. INTRODUCTION

As the deterioration of the electromagnetic environment, the noise in received signal of torpedo magnetic fuse will be more severe and complex. In order to ensure the quality of signal detection and operating reliability of fuze, torpedo magnetic fuze signal denoising is an effective and important technology way. The reference[1] and [2] studied the wavelet denoising method for target signal of torpedo electromagnetic fuze, magnetic fuze and ultrasonic fuze, and certain effect on reducing noise was made. But the wavelet denoising is only to decompose the low-frequency part of the signal, and the high-frequency part is not be subdivided, so the high-frequency area of the useful signal may be filtered out or weakened[3]. As a promotion of wavelet transform, wavelet packet transform (WPT) provides a more precise method for signal decomposition, by multi-level division of the band, the high-frequency part that not be subdivided in discrete wavelet transform is further decomposed. According to the characteristics of the signal being analyzed, the WPT can select the appropriate frequency band adaptively, so as to matching with the signal, and improving the time-frequency resolution. In this paper, wavelet packet method is used for denoising the fuze signal under complex background interference, leading to advance the reliability of torpedo magnetic fuze operation, in order to more effectively ensure the reliability of torpedo magnetic fuze operation.

II. MAGNETIC FUZE SIGNAL MODEL

Taking the target signal of typical torpedo magnetic fuse as signal model to begin the research, and their received target signal can respectively be approximated as Gaussian pulse[4], which can be expressed as follows:

$$H(t) = H_m \exp[-\beta(t_0 - t)^2] \quad (1)$$

Where, H_m is the amplitude of envelope signal, β is a constant, and relative to the parameters of the relative speed, encounter angle, the equivalent width of the target ship and hitting distance.

Empirical formula is expressed as follows[5]:

$$\beta = \frac{0.36\sin^2(\theta_k)V_{TK}^2}{B_k^2 + (1 + 0.063h)} \quad (2)$$

Where, θ_k is encounter angle between torpedo and target, V_{TK} is relative speed of torpedo and target, $V_{TK} = V_T + V_K \cos(\theta_k)$, B_k is equivalent width of the target ship, is hitting distance.

III. BASIC THEORY OF WAVELET PACKET DECOMPOSITION AND RECONSTRUCTION

For a given orthogonal scaling function and its corresponding wavelet function, two scale equations is expressed as follows[6]:

$$\begin{cases} \varphi(t) = \sqrt{2}\sum_n h_0(n)\varphi(2t-n) \\ \phi(t) = \sqrt{2}\sum_n h_1(n)\varphi(2t-n) \end{cases} \quad (3)$$

Where, $h_0(n)$ and $h_1(n)$ are conjugate filters, if $\varphi(t) \to u_0(t)$, $\phi(t) \to u_1(t)$, $h_0(n) \to h_n$, $h_1(n) \to g_n$,so equation (3) can be expressed as

$$\begin{cases} u_0(t) = \sqrt{2}\sum_k h_k u_0(2t-k) \\ u_1(t) = \sqrt{2}\sum_k g_k u_0(2t-k) \end{cases} \quad (4)$$

By the recurrence relation, $u_n(t)$ can be defined as

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$$\begin{cases} u_{2n}(t) = \sqrt{2}\sum_k h_k u_n(2t-k) \\ u_{2n+1}(t) = \sqrt{2}\sum_k g_k u_n(2t-k) \end{cases} \tag{5}$$

The function family $\{u_n\}_n \in z^+$ is called wavelet packet determined by scaling function $u_0(t) = \varphi(t)$ .The formula of wavelet packet decomposition is expressed as follows[7]:

$$\begin{cases} d_{j,n}^{(2l)} = \sum_{m\in Z} h_{m-2n}^{*} d_{j+1,m}^{(l)} \\ d_{j,n}^{(2l+1)} = \sum_{m\in Z} g_{m-2n}^{*} d_{j+1,m}^{(l)} \end{cases} \tag{6}$$

Where, h_k and g_k are wavelet filter banks, h_k is low-pass filter, g_k is high-pass filter.

The above equations indicate that wavelet packet also decomposes the high-frequency components of signal continuously while decomposing the low-frequency components of signal, this is the main differences between wavelet packet decomposition and wavelet decomposition. Wavelet packet decomposition tree is a more intuitive representation for wavelet packet decomposition, and the three-scale wavelet packet decomposition tree is shown as Fig. 1. Wavelet packet decomposition tree allows the signal S to be expressed as S=A+AAD+DAD+DD, where, A is the low-frequency components of the signal, D is the high-frequency components of the signal, the subscripts are the layers of wavelet packet decomposition.

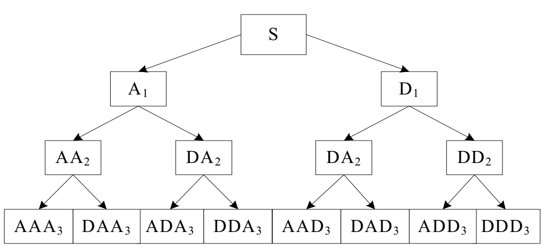


Figure 1. Three-scale wavelet packet decomposition tree.

The instead regression algorithm reconstructed the above signal decomposed by wavelet packet from the bottom to the top, so the reconstruction formula for the wavelet packet can be obtained as follows[7]:

$$d_{j+1,m}^{(l)} = \sum_{m\in z}\left(h_{m-2n} d_{j,n}^{(2l)} + g_{m-2n} d_{j,n}^{(2l+1)} \right) \tag{7}$$

IV. PRINCIPLE OF WAVELET PACKET DENOISING

One-dimensional noisy signal model can be expressed as

$$f(k) = s(k) + \varepsilon \cdot e(k) \tag{8}$$

Where, $f(k)$ is the noisy signal, $s(k)$ is useful signal, in practice, often as low frequency or steady-state, $e(k)$ is noise, usually high-frequency, and contained in the high-frequency components.

The essence of signal denoising is to suppress useless signal, at the same time the useful component be recovered, the signal denoising processes based on wavelet packet can be divided into four-steps as follows:

a) Wavelet packet decomposition of signal: First choose a wavelet function and determine the required level of decomposition, then using wavelet packet to decompose the signal.

b) Determine the optimal wavelet packet basis, then compute the optimal tree the for a given entropy standard.

c) Quantify the threshold of wavelet packet coefficients: For each wavelet packet coefficient, select an appropriate threshold and quantify the threshold of coefficient.

d) Wavelet packet reconstruction of signal: According to the bottom coefficients of wavelet packet decomposition and coefficient that its threshold has been quantified, to reconstruct the wavelet packet.

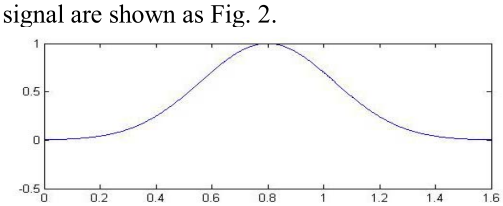
The choice of the threshold is a key point for wavelet packet denoising method, to some extent, it is directly related to the quality of signal denoising. If the threshold is too small, the denoised signal is still noisy, in contrast, if the threshold is too large, some important characteristics of signal would be filtered, leading to deviation. In this paper, the two ways that default threshold denoising and the adjusted threshold denoising are used.

a) Default threshold denoising: This method is the most typical in denoising methods[3]. First obtain the default threshold in the process of denoising, then use the default threshold to denoise the signal.

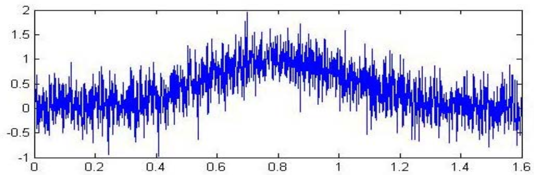
b) Adjust threshold denoising: According to the previous denoising effect, adjust the current threshold. Selecte different thresholds based on your own needs, because the selection of the threshold is related to the denoising effect that is good or bad, so the threshold adjustment is critical.

V. SIMULATION EXPERIMENT

According to equation (1), the parameters of the typical torpedo magnetic fuze signal are selected. Suppose the background noise is Gaussian white noise, and its SNR is 8, using MATLAB to simulate the corresponding waveforms, then the simulation waveforms of magnetic fuze signal and noisy signal are shown as Fig. 2.



（a）useful signal



（b）noisy signal (white noise, SNR=8)

Figure 2. Magnetic fuze signal and noisy signal.

Select the db8 wavelet function to decompose the noisy signal by 8 layers wavelet packet, the denoising results under conditions of default threshold are shown as Fig. 3.

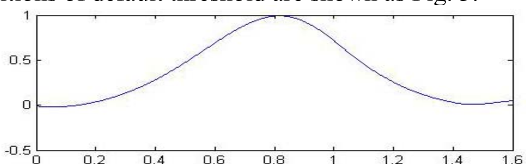


Figure 3. Magnetic fuze signal denoising results in the background of white noise.

Suppose that the background noise is colored noise, denoising in the same way, the results are shown as Fig. 4.

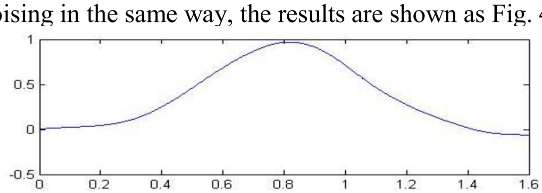
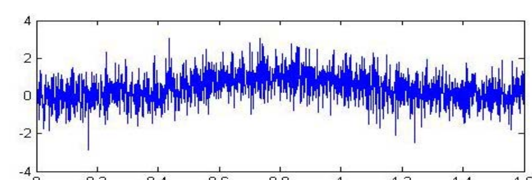


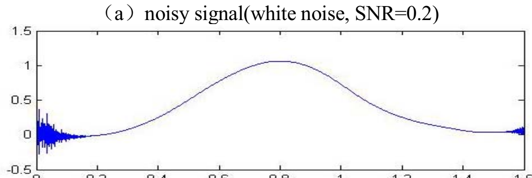
Figure 4. Magnetic fuze signal denoising results in the background of colored noise.

From the Fig. 3 and Fig. 4, it can be seen that for the magnetic fuse signal, when the SNR is large, regardless of background noise is white noise or colored noise, the signal waveform almost no distortion after the default threshold denoising, only when the background noise is color noise, signal amplitude is decreased slightly. Thus, there is no need to adjust the threshold.

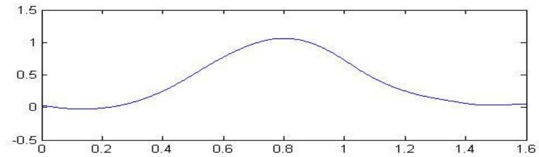
To test denoising effect of wavelet packet in the background strong noise, firstly, soppose the background noise is Gaussian white noise, still select the db8 wavelet function to decompose the noisy signal by eight-layer wavelet packet, thus in conditions that SNR is 0.2, denoising results of wavelet packet are shown as Fig. 5.



（a）noisy signal(white noise, SNR=0.2)



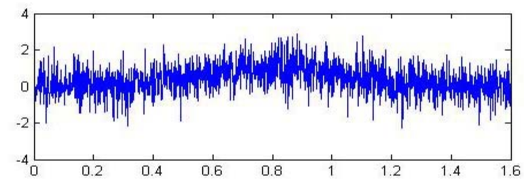
（b）default threshold



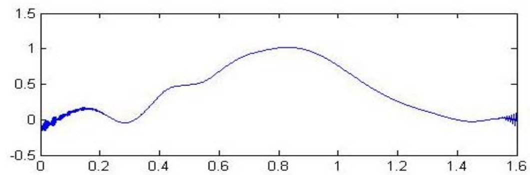
（c）adjust threshold

Figure 5. Magnetic fuze noisy signal and denoising results in the background of strong white noise.

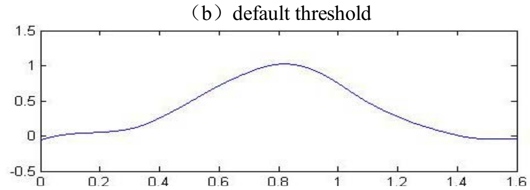
Making the background noise is colored noise, and also in the signal to noise ratio SNR = 0.2 conditions, in this case, the signal is completely submerged in colored background noise. Then select the db8 wavelet function to decompose the noisy signal by eight-layer wavelet packet, the denoising results are shown as Fig. 6.



（a）noisy signal(colored noise, SNR=0.2)



（b）default threshold



（c）adjust threshold

Figure 6. Magnetic fuze noisy signal and denoising results in the background of strong colored noise

According to the above simulation graphs, the following conclusions can be got. When the signal to noise ratio is small, regardless of background noise is white noise or colored noise, the signal is completely submerged in the noise, and all signal waveforms have some distortion after the default threshold denoising, and the waveform distortion in the background of colored noise is larger than in the background of white noise, which shows that colored noise is more difficult to eliminate than white noise, and this is consistent with the theory. But after adjusting the threshold, the waveform distortion disappeared basically. In the study, the denoising effect that using Daubechies wavelet families is better than the others.

VI. CONCLUSIONS

In this paper, wavelet packet transform is used to denoise the torpedo magnetic fuze signal , and the waveform close to the real signal is obtained. Both theoretical and simulation results show that the WPT denoising method can effectively eliminate background noise existed in torpedo magnetic fuze signal, and has the following advantages: high precision, no distortion, retain the original characteristics of the signal, etc., and can be used to extract weak signal in the background of strong noise, so it is an ideal means of magnetic fuze signal denoising. In the process of denoising, how to select the wavelet packet basis and re-adjust the threshold is the key to the denoising quality good or bad, and should combined with the characteristics of the signal to select the wavelet packet basis, the threshold and the layer of wavelet packet decomposition, etc.

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